5. The break statement is optional. If omitted, execution will continue on into the next case. The flow of control will fall through to subsequent cases until a break is reached.

6. Nesting of switch statements is also allowed.

Illustration 10.6 – C++ program to demonstrate switch statement

```
#include <iostream>
using namespace std;
int main()
{
        int num;
        cout << "\n Enter week day number: ";
        cin >> num;
        switch (num)
        {
                case 1 : cout << "\n Sunday"; break;
                case 2 : cout << "\n Monday"; break;
                case 3 : cout << "\n Tuesday"; break;
                case 4 : cout << "\n Wednessday"; break;
                case 5 : cout << "\n Thursday"; break;
                case 6 : cout << "\n Friday"; break;
                case 7 : cout << "\n Saturday"; break;
                default: cout << "\n Wrong input....";
        }
}
```

Output:

```
Enter week day number: 6
Friday
```

10.4.7 Switch vs if-else

"if-else" and "switch" both are selection statements. The selection statements, transfer the flow of the program to the particular block of statements based upon whether the condition is "true" or "false". However, there are some differences in their operations. These are given below:

Key Differences Between if-else and switch

S.No	if-else	Switch
1	Expression inside if statement decide whether to execute the if block or under else block.	expression inside switch statement decide which case to execute.
2	An if-else statement uses multiple statements for multiple choices	switch statement uses single expression for multiple choices.
3	If-else statement checks for equality as well as for logical expression.	switch checks only for equality.
4	The **if** statement evaluates integer, character, pointer or floating-point type or Boolean type.	switch statement evaluates only character or a integer data type.
5	If the condition is false the else block statements will be executed	If the condition is false then the default statements are executed.

Chapter 10 Page 152-179.indd 162 3/24/2020 9:21:33 AM